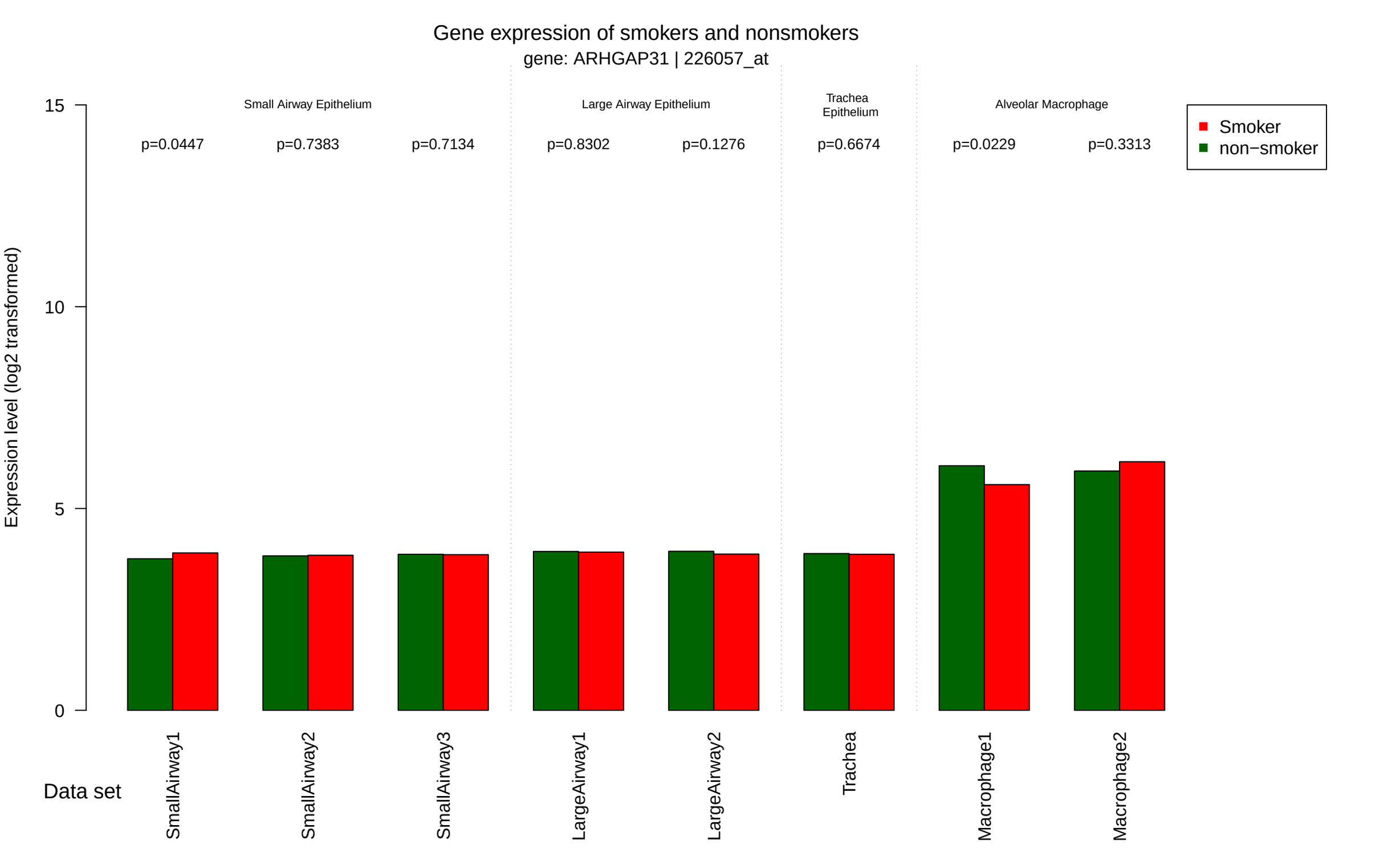
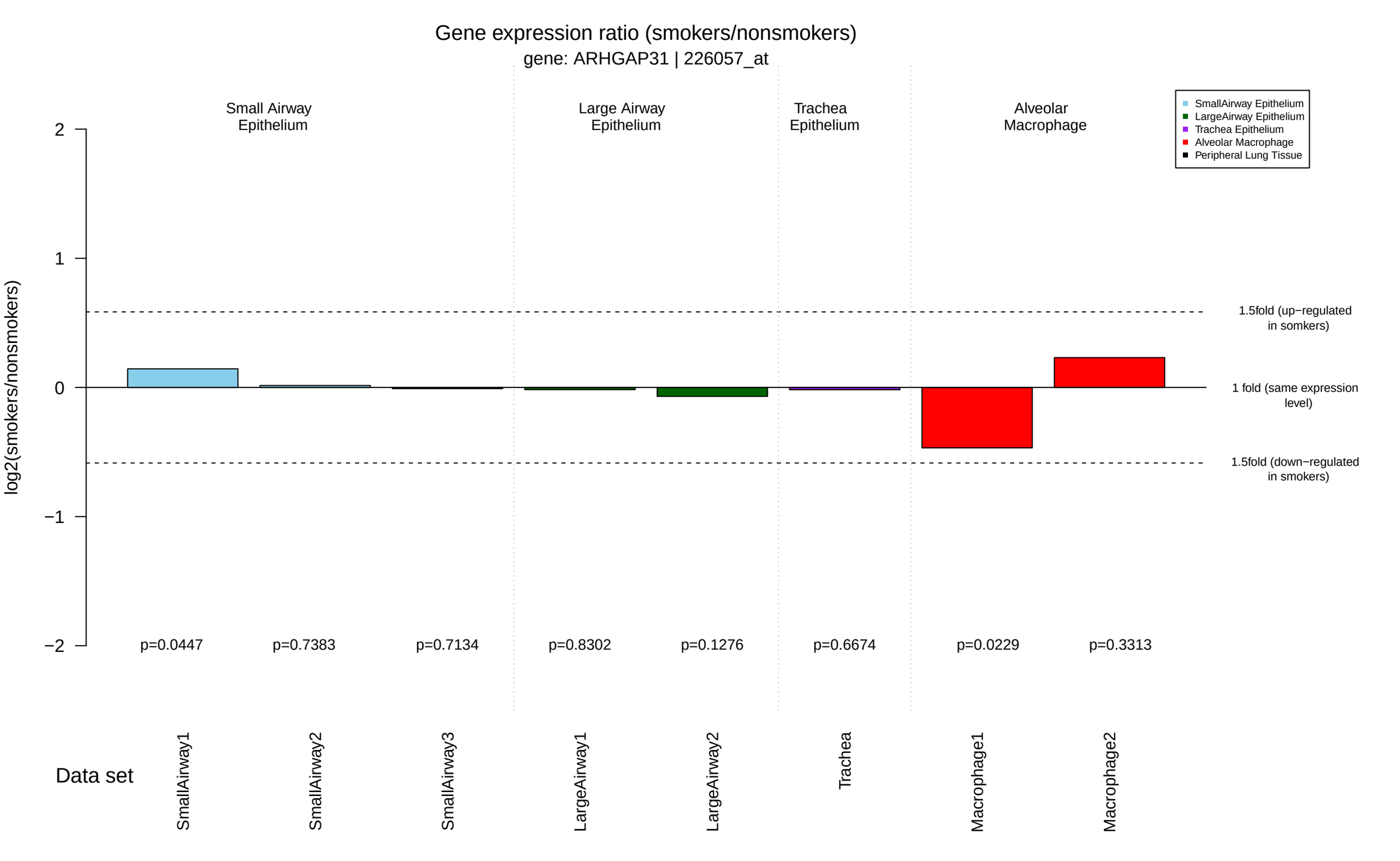
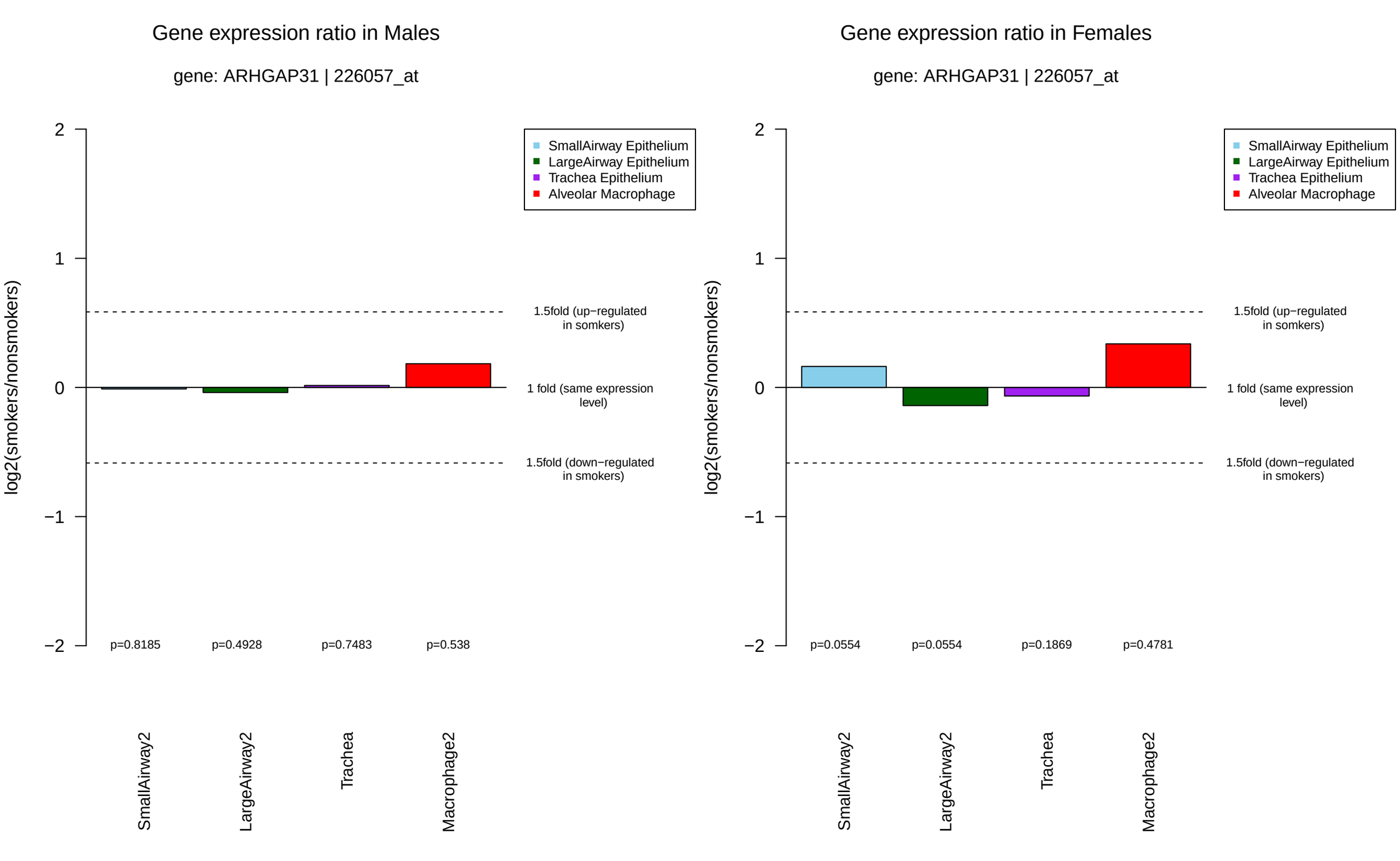
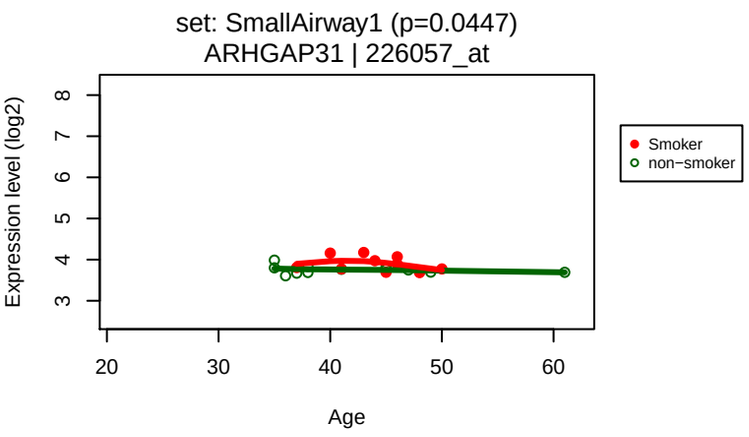
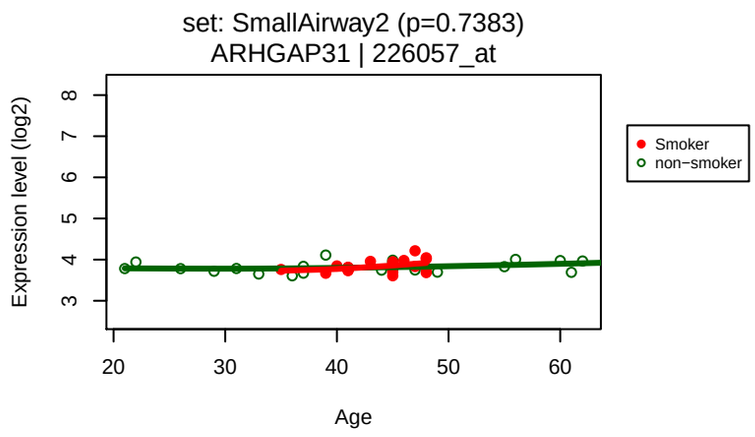
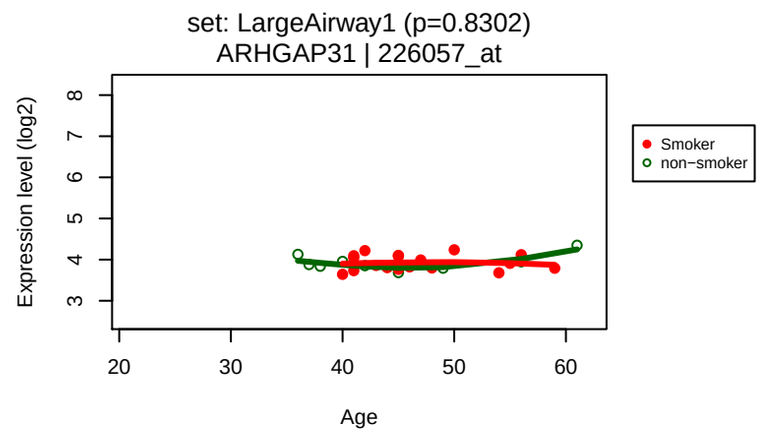

Gene expression of smokers and nonsmokers
gene: ARHGAP31 | 226057_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0447
p=0.7383
p=0.7134
p=0.8302
p=0.1276
p=0.6674
p=0.0229
p=0.3313
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

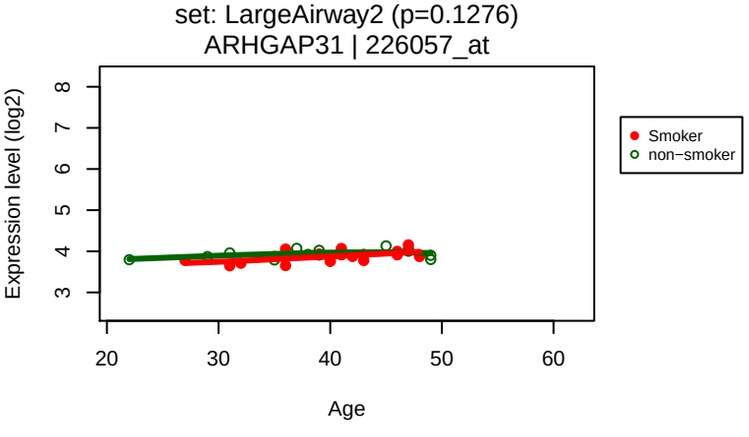

Gene expression ratio (smokers/nonsmokers)
gene: ARHGAP31 | 226057_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0447
p=0.7383
p=0.7134
p=0.8302
p=0.1276
p=0.6674
p=0.0229
p=0.3313
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

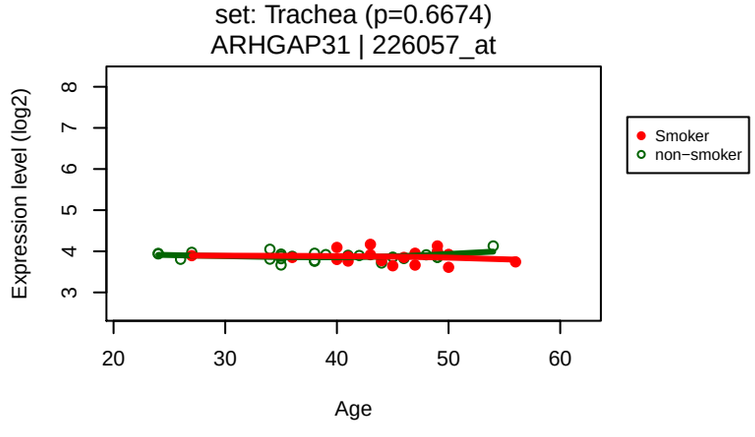

Gene expression ratio in Males
gene: ARHGAP31 | 226057_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8185
p=0.4928
p=0.7483
p=0.538
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGAP31 | 226057_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0554
p=0.0554
p=0.1869
p=0.4781
SmallAirway2
LargeAirway2
Trachea
Macrophage2

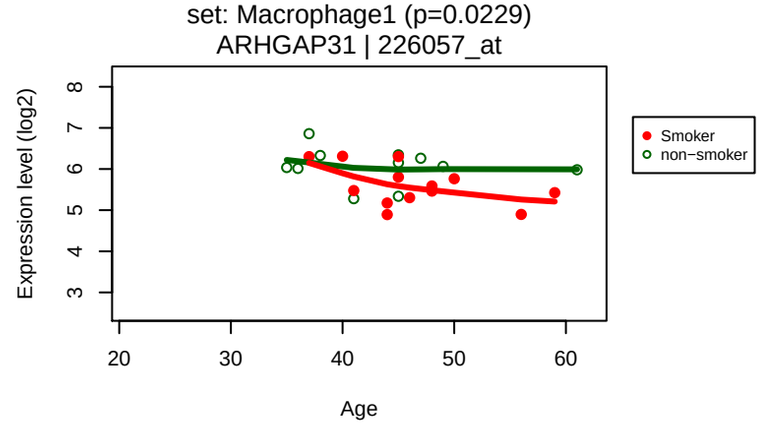

set: SmallAirway1 (p=0.0447)
ARHGAP31 | 226057_at
Expression level (log2)
Age
Smoker
non−smoker

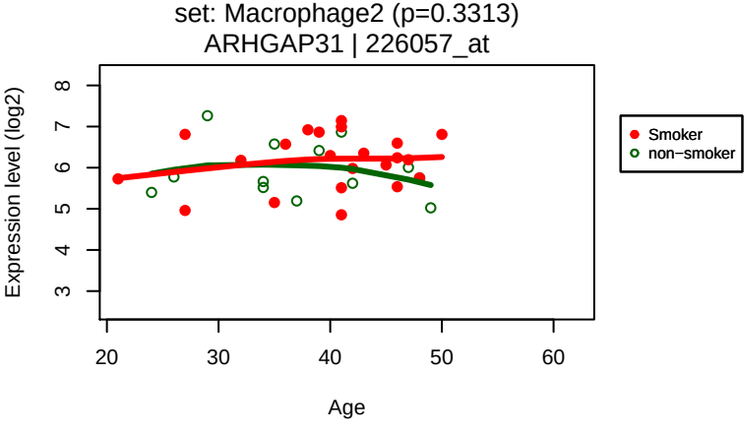

set: SmallAirway2 (p=0.7383)
ARHGAP31 | 226057_at
Expression level (log2)
Age
Smoker
non−smoker

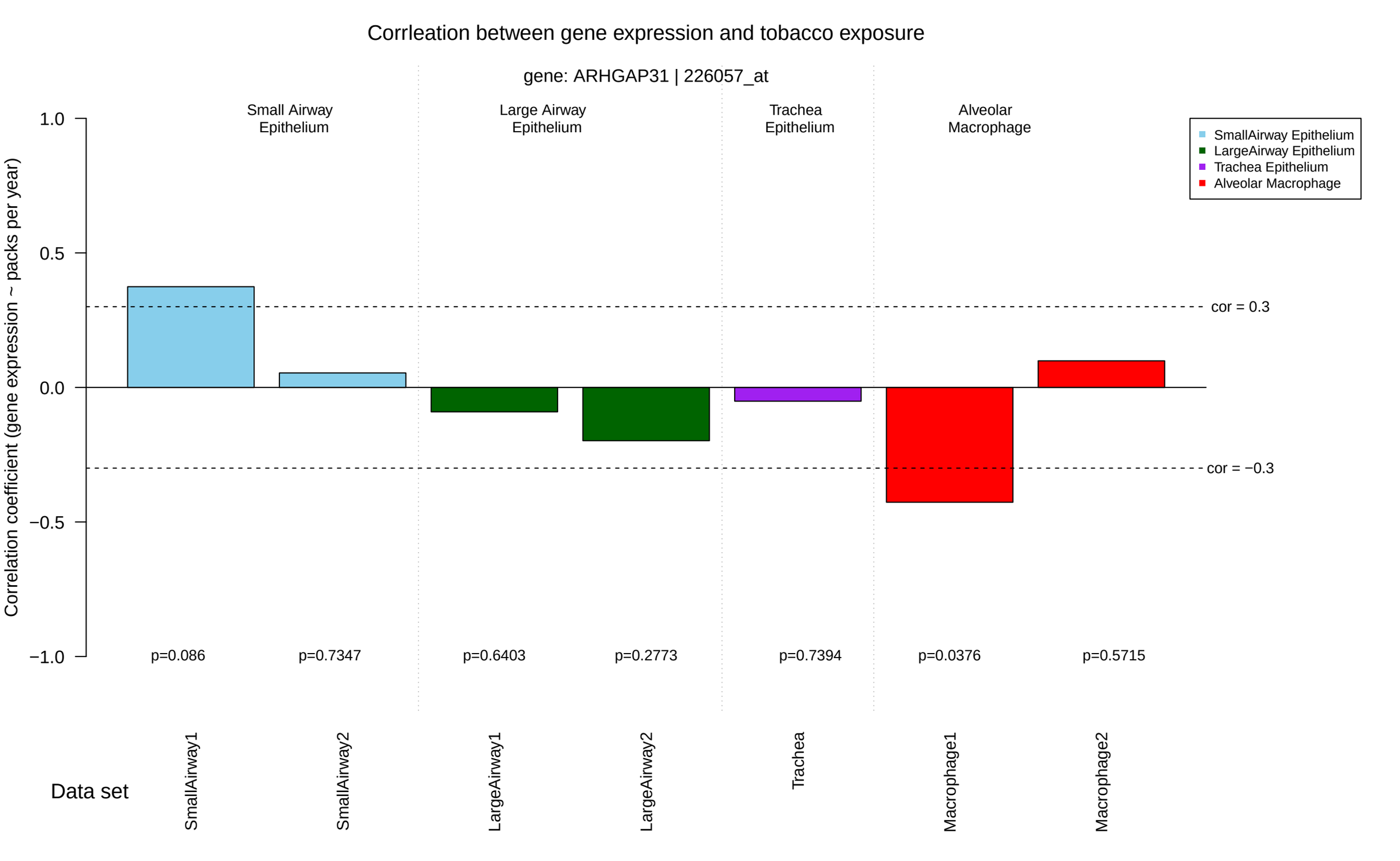

set: LargeAirway1 (p=0.8302)
ARHGAP31 | 226057_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1276)
ARHGAP31 | 226057_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6674)
ARHGAP31 | 226057_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0229)
ARHGAP31 | 226057_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3313)
ARHGAP31 | 226057_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGAP31 | 226057_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.086
p=0.7347
p=0.6403
p=0.2773
p=0.7394
p=0.0376
p=0.5715
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2